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## **Figure 1 Farmers' terms of trade indicators**

Note: GVP = gross value of production

Source: ABARES

## **Summary**

This paper reports on the results of an update to the way ABARES calculates the farmers' terms of trade (FToT) indicators. This includes updates to the price indexes for farm outputs and inputs to production, and measures of farm costs and net farm returns. These indicators are used together with other data and information to monitor the performance of the agricultural sector.

The last major update of the FToT was conducted in 2004–05. This update seeks to address data availability and quality concerns, which have arisen gradually over time. The aim of this update is to improve the accuracy of the indicators and to ensure the ongoing delivery of the FToT indicators. Data sources were reviewed and outputs consolidated where necessary.

The update ensures the estimates are robust and better aligned to the needs of stakeholders. The revised series include significant revisions to farm costs and prices paid indicators, particularly over the most recent 10 years from 2009–10 to 2018–19. Given the many required changes, the updated indicators, which begin in 1994–95 cannot be combined with the former indicators.

Farm costs are aligned with ABS benchmarks and prices paid indicators are updated from various sources. The use of ABS benchmarks align these indicators closer to the Australian System of National Accounts. However, there are differences between ABS and ABARES measures of farm costs, income and net returns, which are outlined in this report.

The revisions to farm costs and prices paid in this update result in an increase of both measures over recent years. This has led to a downward revision to reported net farm cash income for the sector and the FToT index. The updated FToT indicators reveal important insights about the performance of the agricultural sector. For instance, while the value of farm production increased rapidly, net farm cash income for the sector increased at a slower rate because of rising input costs.

This update reflects a broader project investigating farm costs and input prices and provides the basis for ongoing investigation and improvements to the FToT indicators.

## **Introduction**

The availability of data used to compile the FToT has gradually diminished, and data quality concerns have emerged. The cessation of ABS surveys and ABARES collections over time has restricted the amount of data available for the compilation of FToT indicators. This has put constraints on the level of

detail that can be reliably reported. Data quality issues have emerged because the regular benchmarking of farm cost and prices paid indexes ceased in 2004–05. The practice of regular benchmarking is important for checking that the methods used to estimate variables reasonably reflect the real world. This update has reviewed the available data and re-benchmarked the farm costs and prices paid indexes. Changes to the composition of input cost categories have also been made to adapt to data availability.

The farmers' terms of trade (FToT) refers to a suite of indicators that measure average changes to prices and farm costs of the agricultural sector. Price indexes measure the average growth in prices that farmers receive at the farm gate for their product, and in the prices paid for inputs to production. Total farm costs and net returns to the agricultural sector are also estimated. Forecasts of these indicators are produced and updated by ABARES on a quarterly basis.

The headline indicator of the FToT is the FToT index. This index measures changes in the price of outputs compared with the inputs used in their production. Specifically, it is the ratio of the prices received index and the prices paid index. A declining FToT index means that growth in input prices paid by Australian farmers is rising faster than the growth in prices received for their products at the farm gate.

The prices received index is a weighted average measure of the change in the farmgate price of agricultural products. Prices received indexes are also available by commodity group, such as 'crops' and 'livestock'. The prices paid index is a weighted average measure of input prices. The price indexes are available by input type, such as chemicals, feed and fuel.

Estimates of farm costs are available by input type. Measures of the net return to farm production are derived by subtracting farm costs and depreciation from the gross value of agricultural production.

This paper begins with a discussion on the relevance of the FToT (Chapter 2) before detailing how the update was conducted, and why ABS benchmarks were adopted as the measure of farm costs (Chapter 3). In Chapter 4 the results of the update are presented and compared with the former indicators. Farm costs are evaluated first, and the difference between ABARES and ABS measures of income and output are explored. Lastly, the revisions to prices paid and prices received and their implications for the FToT are discussed.

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